

Erdős Problem #635

EP-635

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $t \geq 1$ and $A \subseteq \{1, \dots, N\}$ be such that whenever $a, b \in A$ with $b - a \geq t$ we have $b - a \mid b$. How large can $|A|$ be? Is it true that $|A| \leq \left(\frac{1}{2} + o_t(1)\right)N$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>